

$$T = 625 K, \quad t = 500 \text{ days} \quad t_c = 600 \mu m$$

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$$\mu = 10^{-5} m$$

$$cm = 10^{-2} m$$

$$a) \quad t^*(d) = 6.42 E^{-7} \exp\left(\frac{11949}{625}\right) = 133.0 \text{ days}$$

$$t > t^*$$

$$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{625}\right) = 2.115 \mu m$$

$$K_L = 7.48 E 6 \exp\left(\frac{-12500}{625}\right) = 0.0154 \left(\frac{\mu m}{d}\right)$$

$$\rightarrow \delta(\mu m) = 2.115 + 0.0154 (500 - 133.0)$$

$$\boxed{\delta_{Ox} = 7.77 \mu m}$$

(from above)

$$b) \quad f_H = .14, \quad \delta_{Ox} = 7.77 \mu m, \quad \rho_{Ox} = 5.68 g/cc, \quad \rho_{met} = 4.5 g/cc$$

$$c) \quad f_{ZrO_2}^0 = \frac{2.16}{2.16 + m_{Zr}} = 0.26 \quad \frac{m_H}{m_O} = \frac{1}{14} = .0714 \quad t_c = 600 \mu m$$

$$\rightarrow C_H^{clad} = \frac{2 \cdot (0.14) \cdot 7.77 \cdot 5.68 \cdot 0.26 \cdot 0.0714}{(600 - \frac{7.77}{1.54}) \cdot 4.5}$$

$$= \frac{0.2}{3867.4} \times 10^4 = \boxed{51.7 \text{ wt. ppm}}$$

5) may mitigate reactivity along with negative temp and void coeffs, decay heat and buildup of rod internal pressure can lead to plasticity events in ballooning and bursting.
- still need more

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6) Burnup impacts the specific enthalpy and total displacements via knock on events. Higher burnup materials tend to show preference for a sudden failure due to a material becoming less elastic (accumulation of defects). During an accident, this can lead to premature fracture and cladding failure. Higher burnup fuel also has a lower T_{melt} and K_{th} , but those are less important.
- almost all

4/6
7) limiting corrosion, limiting hydrogen pickup, increasing margins to thermal limits, increasing rad. damage resistance, and fuel clad chemical and mechanical interactions all make a fuel sys. more acc. tolerant.
- yes, but these fall into four general areas that are broader

2/6
8) One near term ATF concept of interest is the use of MOX. This introduction of Pu into UO_2 displays qualities of "burnup resistance" in its fission formation, ability to recycle weapons Pu, and ability to be flexibly introduced into LWR cores with UO_2 , however, typically is in Sol. Fast reactors.
- MOX not ATF

5/6
9) When exposed to high T steam, Zr cladding oxidizes at a much faster rate. T_s in the $1200 > T$ may allow for oxidation of the cladding entirely. Remember diffusion of O is impacted by T_{clad} . -worst case

10) Three examples of ~~the~~ limiting LWR phenomena are
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- margins to thermal limits
- normal operating conditions
- cladding corrosion

11) CRUD lowers the thermal efficiency of the fuel clad system (lowers heat trans. to water) and its constituents may become activated and deposit themselves within the reactor system, increasing rad. exposure to plant operators.

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12) Two water chem. controls implemented in LWRs are the introduction of

- H_3BO_3 (Boric Acid)
 - these are coupled as part of the same thing
 - used as reactivity shim in PWRs
- $LiOH$ (Lithium hydroxide)
 - strong base that retain a nominal pH and allows for excess H to impact coolant (O potential) $\rightarrow$ environment

13) unique MOX microstructures

• JOU (French words) ✓

• a squishy composite of fission products on the outer fuel surface that may act as mechanical buffer to clad

• Annulus ✓

• central void present in MOX fuel
bc steep ∇T induces vapor transport up ~~through~~ ∇T , leaving radial paths of hydride and columnar grains